

ELEC 537: Intelligent Mobile Systems

Instructor Information

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Course Overview

This course focuses on building intelligent mobile systems that sense, communicate, and make decisions under tight resource constraints like limited energy, compute, and bandwidth. Students will explore wireless communication, embedded sensing, and machine learning techniques for mobile applications. Topics include wireless localization, sensor fusion, on-device AI, energy harvesting, wireless imaging, and emerging techniques like 3D mapping and neural radiance fields. The course combines both the foundational principles and the state-of-the-art research in mobile computing and embedded intelligence.

Student Learning Outcomes

By the end of the course, students will be able to:

- Propose, design, and prototype an intelligent mobile system
- Analyze and implement wireless sensing techniques
- Design energy-efficient embedded AI systems for resource-constrained platforms
- Evaluate trade-offs between computation, communication, and energy in mobile systems
- Apply machine learning techniques to wireless sensing and mobile computing problems

Prerequisites/Recommended Background

Signal processing, wireless communication, embedded systems, networking, and machine learning.

Assessment Structure

- Assignments (25%) - 2–3 assignments
- Midterm (25%) - One open-book midterm exam
- Presentation (10%) - Student paper presentation
- Project (40%) - One semester-long group project (teams of 2–4)

Project Description

Students will propose, design, and prototype an intelligent mobile system as part of a group project. Projects will focus on wireless sensing, energy-efficient embedded AI, and/or distributed learning on resource-constrained platforms. The project encourages creativity and innovation at the intersection of mobile computing, embedded AI, and networking.

Example themes:

- RF-based localization
- Wireless imaging and 3D reconstruction
- Acoustic communication underwater
- Battery-free embedded inference
- Federated learning in IoT
- Side channel attacks and security

Weekly Topics and Lecture Schedule

Week 1 [Introduction]

- Lecture 0.1: Course overview, expectations, and applications of mobile/wireless AI systems

Week 2 [Localization]

- Lecture 1.1: Fundamentals of RF-based localization: ToF, AoA, and fingerprinting
- Lecture 1.2: Channel State Information (CSI) and RF-based motion tracking

Week 3 [Localization]

- Lecture 2.1: Learning-based localization and semantic mapping
- Lecture 2.2: Multi-modal sensor fusion: RF, IMU, and vision

Week 4 [Embedded AI & Edge Inference]

- Lecture 3.1: TinyML, model compression, and embedded inference
- Lecture 3.2: Intermittent computing and energy-aware execution models

Week 5 [Battery-Free Systems]

- Lecture 4.1: Energy harvesting from RF, solar, and motion
- Lecture 4.2: Battery-free sensing and computing platforms

Week 6 [Wireless Communication]

- Lecture 5.1: Wireless PHY/MAC fundamentals and digital modulation
- Lecture 5.2: DFT and spectral analysis for wireless signal processing

Week 7 [Wireless Sensing & Radar]

- Lecture 6.1: FMCW radar principles and applications in sensing
- Lecture 6.2: RF sensing for activity recognition and gesture tracking

Week 8 [Acoustic Sensing]

- Lecture 7.1: Basics of acoustic propagation and time-of-flight sensing
- Lecture 7.2: Ultrasound and audio-based positioning and interaction

Week 9 [Edge & Federated AI]

- Lecture 8.1: Edge-cloud coordination and pipeline partitioning
- Lecture 8.2: Federated learning and communication-efficient training

Week 10 [Security & Privacy]

- Lecture 9.1: Spoofing, jamming, and threats in localization and sensing
- Lecture 9.2: Privacy in mobile sensing and federated systems

Week 11 [Wireless Imaging & NeRF]

- Lecture 10.1: Neural radiance fields (NeRFs) and their use in wireless vision
- Lecture 10.2: Gaussian splatting for RF scene reconstruction

Week 12 [Smart Surfaces & Metamaterials]

- Lecture 11.1: Intelligent reflecting surfaces (IRS) and programmable wireless
- Lecture 11.2: Metamaterials for wireless manipulation and sensing

Week 13 [Distributed Systems & Applications]

- Lecture 12.1: Swarm intelligence and distributed mobile AI systems
- Lecture 12.2: Applications to infrastructure, health, and climate sensing

Week 14 [Project Presentations]

- Lecture 13.1: Final project presentations (Part 1)
- Lecture 13.2: Final project presentations (Part 2)

Grade Policy

Final grades will be calculated based on the weighted average of all course components as outlined in the Assessment Structure. Late assignments will be penalized 10% per day unless prior arrangements are made with the instructor. Extensions may be granted for documented emergencies or health issues.

Workload Statement and Breakdown

The course includes three assignments. The first will be completed individually and released in Week 3. The second and third will be done in groups of two and released in Weeks 6 and 9, respectively. These assignments will cover key technical topics such as wireless sensing, localization, and embedded AI. Together, the assignments account for 25% of the final grade.

An open-book midterm exam will be held in the last week of October (Week 10). It will test students on material from the first half of the course, including foundational principles and system-level thinking. The midterm will account for 25% of the final grade.

Each student will also present one state-of-the-art research paper during the semester. Presentations will be evaluated on clarity, technical depth, and the ability to lead class discussion. This component is worth 10% of the final grade.

The course includes a semester-long group project (teams of 2–4 students). Students will design and prototype a mobile AI system with components in sensing, communication, or embedded intelligence. A list of suggested project topics will be provided, though students are encouraged to propose original ideas in consultation with the instructor. A progress report will be due in Week 8, and final reports and presentations will be due in Week 14. The project accounts for 40% of the final grade.

Workload Breakdown Table (Total: 84 hours):

Activity	Estimated Hours	Grade Weight
Assignment 1 (individual)	7 hours	8.33%
Assignment 2 (group)	7 hours	8.33%
Assignment 3 (group)	7 hours	8.33%
Midterm exam (prep + completion)	15 hours	25%
Student paper presentation	6 hours	10%
Final project (progress + final report and presentation)	42 hours	40%
Total Estimated Student Workload	84 hours	100%

Course Policies and Resources

Rice Honor Code

All students must uphold the Rice Honor Code standards established when you matriculated at Rice. If you need clarification on academic integrity expectations, procedures, or student rights, consult the Honor System Handbook at <http://honor.rice.edu/honor-system-handbook/>. This resource details what constitutes academic misconduct and outlines the resolution process.

Disability Resource Center

Any student with a documented disability needing academic accommodations should: 1) visit the Disabilities Resource Center (DRC) to ensure required documentation is on file, and 2) speak with the instructor as soon as possible. The DRC is located in Allen Center 111 and can be reached at adarice@rice.edu or x5841.

Mental Health Statement

Student wellbeing and mental health are important. If personal challenges are interfering with your ability to complete coursework, please contact the Wellbeing and Counseling Center for support. Rice offers free mental health services through the Wellbeing and Counseling Center to help students navigate personal difficulties that may impact their academic or personal wellbeing. Students experiencing significant stress, sadness, or anxiety can receive assistance from the Student Wellbeing Office or Rice Counseling Center. The Wellbeing and Counseling Center operates from the Gibbs Wellness Center and provides 24/7 support at 713-348-3311.

Updates to the Course

Information in this syllabus may be subject to change with reasonable advance notice as appropriate.